

# Research on intelligent scene optimization of film and television director based on NSGA-III algorithm

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## ABSTRACT

In the context of the vigorous development of the global film and television industry and the deep integration of creativity and technology, traditional film and television scene design methods can no longer meet the growing and complex needs of the industry because they struggle to balance artistic quality with production efficiency. In light of this, this study makes a breakthrough by applying the NSGA-III algorithm to the field of film and television scene optimization, achieving the best balance between scene aesthetics and functionality through the construction of a multi-objective optimization strategy. This effectively solves the problem of multi-rights protection and balance in film and television scene design. In 10 classic film and television scene optimization experiments, the scheme generated by the algorithm increased the visual effects score by 12 %, the plot fit by 9 %, and reduced costs by 15 %. At the same time, with its efficient computing power, the original scene design cycle of several weeks has been greatly shortened to several hours, significantly saving manpower and material resources. In addition, the results of this research were successfully demonstrated in the pilot of commercial projects across three different thematic genres (action, sci-fi, and costume), reducing the construction cycle by an average of 20 %, increasing the material reuse rate by 18 %, and reducing the number of reworks due to design flaws by 30 %. This research not only provides an innovative technical path for the optimization of film and television scenes but also opens up a new direction for the film and television industry to achieve a win-win situation of artistic innovation and economic benefits. It fills the gap of technology application in this field and has important theoretical and practical value.

## 1. Introduction

With the vigorous development of global cultural and creative industries, film and television creation is absorbing cutting-edge scientific and technological achievements at an unprecedented speed to achieve a qualitative leap in narrative techniques, visual presentation, and audience interaction. As a critical link in the film and television production process, scene design not only carries the critical mission of shaping the story atmosphere and strengthening the character of the character but also directly displays the director's creativity and technical strength [1, 2]. However, in the face of increasingly complex shooting conditions, diversified aesthetic needs, and limited time budget constraints, traditional artificial scene planning methods gradually show limitations, and it is urgent to seek more efficient and intelligent solutions [3].

Non-dominated Sorting Genetic Algorithm III (NSGA-III) algorithm, with its excellent multi-objective optimization ability, has shown extraordinary strength in solving complex problems in many fields, such

as engineering design and economic dispatch [4,5]. The core advantage of this algorithm is that it can consider multiple conflicting objective functions simultaneously, find a set of non-inferior solutions (Pareto optimum), and then provide rich alternatives for decision-makers [6]. For film and television scene optimization, this meets the balanced demands of the triple standards of aesthetics, practicality, and cost-effectiveness in the scene design process [7,8]. On the one hand, the director hopes the scene can accurately convey the script's intention and create a specific emotional atmosphere [9]. On the other hand, the production team needs to consider the difficulty of shooting, the possibility of post-production special effects synthesis, and the rationality of the overall budget [10,11]. Under such complicated needs, the intervention of the NSGA-III algorithm has undoubtedly found the key to solving the dilemma [12].

The application of film and television scene optimization also encounters numerous challenges. For instance, how can subjective aesthetic indicators be accurately quantified, and how can we ensure

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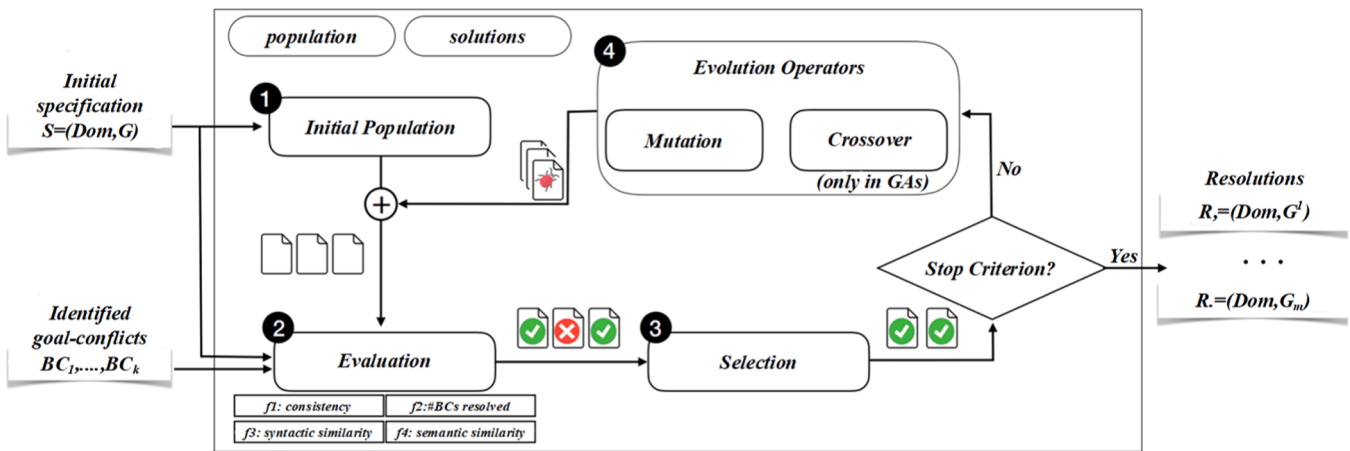


Fig. 1. Schematic diagram of NSGA-III.

that the artistic standards understood by machines closely resemble human sensory experiences? How can cultural background and historical origin be embedded in algorithm design so that the scene is beautiful and rich in deep meaning? In addition, the data collected was more difficult. How can the workload be reduced in the preliminary preparation stage while ensuring the scene's authenticity and the details' richness

The major contribution of this research is the innovative application of the NSGA-III algorithm to the field of intelligent scene optimization for film and television directors, breaking the limitations of traditional scene design that relies on experience and subjective judgment [13,14]. Given the complexity of multi-objective optimization in film and television scene design, a unique set of problem modeling and algorithm adaptation schemes was proposed. This successfully incorporated multiple key objectives, such as visual effects, plot fit, and cost control, into the optimization system [15]. Through extensive practical case verification, the algorithm has been shown to improve the visual effect score of film and television scenes by 12 %, increase plot fit by 9 %, enhance cost control by 9 %, shorten the average scene construction cycle by 15 %, boost the material reuse rate by 15 %, and reduce the number of reworks due to design defects by 15 %. The novelty of this research is evident not only in the development of algorithm application scenarios but also in the construction of a new paradigm that deeply integrates film and television art creation with intelligent algorithms. This provides a reusable technical path and theoretical support for the intelligent transformation of film and television production, filling a gap in the research of multi-objective intelligent optimization in related fields.

Against this backdrop, this study focuses on the potential application of NSGA-III in intelligent scene optimization for film and television directors [16]. It explores an innovative path that integrates creative inspiration with rational decision-making. Specifically, within the film and television industry, this study envisions applying the NSGA-III algorithm to intelligent scene optimization. Initially, it involves the parametric description of scene elements (such as props, lighting, color, layout, etc.) and constructs a mathematical model with multiple variables. Then, leveraging the algorithm's powerful search capabilities, the optimal solution is explored within the defined design space, resulting in a series of scene design schemes that consider visual impact, narrative logic coherence, and economic benefits. To ensure the practicability and operability of the scheme, it is essential to integrate the opinions and experiences of industry experts, fine-tune and optimize the algorithm's results, and ultimately form an execution plan that aligns with the original intention of artistic creation while effectively controlling costs [17,18].

## 2. NSGA-III algorithm related techniques

### 2.1. NSGA-III algorithm

NSGA-III is derived from the NSGA-II framework and introduces a reference point guidance mechanism. NSGA-II promotes the continuation of high-quality genes through non-dominated sequencing and embedding elite inheritance to maintain crowding and ensure diversity and dispersion [19,20]. With the increase of targets, the high-dimensional operation of NSGA-II surges, and the congestion calculation above three dimensions is complicated, which leads to the sharp increase of non-dominated solutions, seizes Pareto resources, and affects the convergence efficiency of the algorithm [21,22].

The schematic diagram of NSGA-III is shown in Fig. 1. NSGA-III maintains good population diversity when facing targets above three dimensions, and the Pareto frontier is more uniformly distributed [23]. Compared with the previous generation algorithm, NSGA-III has made several improvements in the mechanism design. NSGA-III adopts the elite retention strategy guided by reference points, which guides the population evolution direction through the predefined distribution of reference points, effectively avoids blindness and randomness in the search process, and enhances the global search ability [24,25]. Secondly, the algorithm introduces a diversity maintenance mechanism. It uses crowding distance and neighborhood to automatically adjust the population distribution density to ensure the comprehensive coverage of solution space and prevent the constraint of local optimal traps. NSGA-III supports dynamically adjusted cross-mutation probability, which can optimize parameter selection in real-time according to the characteristics of the current solution set, further accelerating the convergence rate and improving the optimization efficiency [26,27].

In order to adapt to more diversified application scenarios, the NSGA-III algorithm can estimate the feasibility of specific solutions in advance by combining prediction models with deep learning, reducing unnecessary computational overhead. Fuzzy logic processes uncertain information to enhance the algorithm's robustness in incomplete or ambiguous data environments. At the same time, through the distributed computing framework, parallel processing in the background of big data is realized to improve the ability to solve large-scale problems.

### 2.2. Intelligent scene of film and television director

The purpose of this part is to lay the research foundation and core background for the full text, and to clarify the scope and connotation of the research object.

In traditional film and television production, scene design often depends on the experience and intuition of directors and art directors.

Every step is full of manual traces and personal style, from the conception of sketches to physical construction. Although this method can endow the works with unique charm, it is difficult to eliminate the problems of excessive subjectivity, low efficiency, and waste of resources [28,29]. Through the intelligent system, the script's content can be quickly analyzed, key plot clues and emotional atmosphere can be automatically extracted, and diversified scene ideas can be generated accordingly, covering all-around details such as layout, light and shadow, and color [30]. Data-based scene optimization can significantly broaden the breadth and depth of creativity and ensure a high degree of compatibility between scene design and storyline, thus improving the overall artistic quality of the work and the audience's immersion experience [31].

Intelligent scene technology has extremely high practical value and economic benefits [32]. By establishing a huge material database, the intelligent system can realize the intelligent retrieval and reuse of resources and significantly reduce physical materials' consumption and transportation costs. At the same time, with the help of virtual reality technology, film and television workers can preview and modify the scene in real-time on the computer screen, thus eliminating the tedious process of repeatedly building physical models and significantly improving creative efficiency [33,34]. In addition, the intelligent scene can also be seamlessly connected with the remote collaboration platform, allowing artists worldwide to share design concepts in real-time and collaboratively improve every detail, breaking geographical restrictions and promoting creative exchanges and integration worldwide [35,36].

### 3. Intelligent scene optimization of film and television director based on NSGA-III algorithm

#### 3.1. Multi-objective optimization design of film and television scenes based on NSGA-III algorithm

In order to adapt the problem to the NSGA-III algorithm, the multi-objective attributes of film and television scene optimization were first clarified, and the core objectives such as visual effects, plot fit, and cost control were transformed into quantifiable mathematical indicators, and an optimization model framework that met the requirements of the algorithm was constructed. The dataset used covers 10,000 scenes from 500 different styles of film and television productions, including video footage, plot text, professional ratings, and cost data. Taking the indoor dialogue scene of a feature film as an example, the feature selection process is demonstrated: from the 30 features initially extracted, such as color saturation and lens switching frequency, 20 effective features are retained by variance filtering, and then 12 key features strongly correlated with the target are screened out by mutual information method.

For computer vision technology, the image segmentation model based on deep learning is used to extract the contours and color distribution of objects in the scene, and the color contrast (for example, the channel difference is calculated after converting the RGB value into HSV space), the light and shadow level (the pixel brightness distribution is analyzed through the grayscale histogram) and other features are calculated by combining the OpenCV library, and the numerical score is mapped to 0-10, for example, the color contrast of a night scene is calculated to score 6.8 points. In terms of semantic analysis, the BERT model is used to extract emotional tendencies and keywords from the plot text, and the semantic information such as "tense atmosphere" and "warm scene" is converted into quantitative vectors, such as the semantic vector value of the chase scene of an action movie is [0.8, 0.1, 0.3], to ensure that the whole research process can be reproduced.

This part aims to introduce the NSGA-III(Non-dominated Sorting Genetic Algorithm III) algorithm to build a scientific multi-objective optimization model for multiple interrelated and sometimes conflicting goals in film and television scene design, such as visual effects, plot fit, and cost control [37]. Through the mechanisms of non-dominant

ranking and congestion calculation of the algorithm, the optimal solution or Pareto frontier solution set that can balance each target is searched in the complex solution space, so as to provide diversified and high-quality scene design solutions for film and television directors.

This study is systematically and comprehensively compared with the traditional artificial empirical design and the scenario design scheme based on the single-objective optimization algorithm. In comparison with traditional manual experience design, the traditional method often relies on the designer's personal subjective judgment and past experience, which is prone to the problem of one-sided optimization direction, such as focusing only on visual effects and ignoring cost control, or sacrificing the efficiency of scene construction to fit the plot [38]. The NSGA-III algorithm-based method can take into account multiple key objectives such as visual effects, plot fit, cost control, and construction cycle at the same time by virtue of its multi-objective optimization characteristics, which effectively avoids the limitations of traditional methods and significantly improves the comprehensive quality of film and television scenes. Compared with the scheme based on a single-objective optimization algorithm, the latter can usually only focus on a specific goal for optimization, and it is difficult to achieve balance and coordination between multiple objectives. With the help of mechanisms such as non-dominant sorting and congestion calculation, the NSGA-III algorithm can efficiently search in the complex solution space, which can not only find more diverse feasible solutions, but also form a broader and balanced Pareto frontier solution set, providing film and television directors with rich and high-quality scene design options to meet the needs of different projects. However, there are limitations to this approach. In the process of practical application, because film and television scene design involve many abstract artistic indicators, such as the creation of visual atmosphere, the expression of plot emotion, etc., it is not easy to accurately quantify these abstract contents and integrate them into algorithm modeling, which undoubtedly increases the complexity and difficulty of problem modeling. At the same time, the NSGA-III algorithm has high requirements for computing resources when processing large-scale scene optimization tasks, and the operation process often requires strong hardware support, and the computing time is long, which limits its rapid application in time-critical projects to a certain extent.

Multi-objective optimization involves optimizing multiple sub-objectives simultaneously. Methodologically, it is divided into normalization of single-objective (such as objective programming and linear weighting) and non-normalization technology that keeps multi-objective characteristics. The non-normalization method is based on Pareto optimality, which avoids the problematic weight-setting problem. Intelligent optimization algorithms, such as non-dominated sorting genetic algorithms based on elite strategy, maintain the diversity of solutions through crowding degree. At the same time, NSGA-III introduces reference point sorting on this basis to further optimize population diversity. In two-dimensional scenarios, NSGA-II (Non-dominated Sorting Genetic Algorithm II) prefers to select individuals with large crowding distances in the non-dominated layer. However, in the face of a three-dimensional lightweight design, the non-dominance between targets leads to the surge of optimal solutions, slows down the algorithm's convergence, and the solution distribution is concentrated, making it difficult to adapt to the screening of two-dimensional features. Therefore, this paper adopts NSGA-III to achieve the goal of being lightweight by taking advantage of its global solid search and low cost of a multi-objective solution.

In this study, the accurate control and efficient deployment of scene design elements are realized by introducing intelligent scenes and using algorithms to optimize scenes. The initial population should have diversity, uniformity, and feasibility to ensure that random individuals are generated under the constraints and the best one is selected from them to form the initial population [39]. The constructed multi-objective optimization model of film and television scenes is combined with NSGA-III, initializes parameters such as reference point, iteration times, and

probability values, and sets the initial generation times to 0. Integer coding associates' genes and performances, and comprehensive random and heuristic methods generate high-quality initial populations.

For the randomly generated initial population  $P_t$ , using Simulated Binary Crossover and Polynomial Variation strategies, the better individuals are selected, and the offspring population  $Q_t$  is generated. The population sizes of  $P_t$  and  $Q_t$  are both  $N$ . If the parent population and the offspring population are merged, and a new population  $R_t$  is formed through their joint competition, the new population size is  $2N$ , that is,  $R_t = P_t \cup Q_t = 2N$ . Constructing a new population  $S_t$ . Based on Pareto non-dominated ranking,  $R_t$  is formed into several different non-dominated layers ( $F_1, F_2, F_3, \dots$ ). Starting from  $F_1$ , the individuals in each non-dominated layer are put into  $S_t$  in turn until  $S_t \geq N$  so that the optimal individuals will not be lost and the diversity of the population in the target space is maintained. Unlike the NSGA-II algorithm, the core idea of the NSGA-III algorithm is an individual selection mechanism based on reference points, which can reduce the computational cost while maintaining population diversity. The specific steps are as follows:

Das and Dennis's reference point method was selected to upgrade the individual screening from two-dimensional to three-dimensional and enhance the efficiency of the non-dominated MOP solution. According to formula (1), points are arranged on the  $(M-1)$ -dimensional hyperplane to ensure the wide distribution of points and improve the diversity of screening.

$$P = \begin{pmatrix} M+H-1 \\ H \end{pmatrix} \quad (1)$$

$P$  is the total number of reference points,  $M$  is the optimized target dimension, and  $H$  is the average value of the number of points taken for each dimensional variable. This paper is an optimization problem of a cab three-dimensional objective, and the distribution of the reference point set is similar to a triangle.

Construct ideal points. By calculating the minimum value among  $M$  optimization objective functions, the ideal point set composed of population  $S_t$  in the three-dimensional objective space is determined, and the expression is (2):

$$Z_i^{\min} = (Z_1^{\min}, Z_2^{\min}, Z_3^{\min}, \dots, Z_M^{\min}) \quad (2)$$

Where:  $Z_i^{\min}$  is the first minimum value of the  $i$ -dimensional target. To realize the translation transformation of the target space and translate the required objective function, the expression is (3):

$$f_i'(x) = f_i(x) - Z_i^{\min} \quad (3)$$

$f_i'(x)$  is the calculation function, which calculates the extra points of the target axis. The additional points on each target axis are obtained by finding the solution that minimizes the following functions, and the transformation weights are axial directions, which can be expressed as (4):

$$A_{SF}(x, w) = i = 1, \dots, M \frac{f_i'(x)}{w}, x \in S_t \quad (4)$$

Where:  $A_{SF}(x, w)$  is the additional points on the target axis;  $x$  is the individual in the population  $S_t$ ;  $w$  is the conversion weight. Build a linear hyperplane. The linear hyperplane is constructed using the extreme vector, and the intercept between the  $i$ -th dimensional target axis and the linear hyperplane is calculated. The linear hyperplane equation can be expressed as (5):

$$A_1x_1 + A_2x_2 + A_3x_3 + \dots + A_Ux_U = 1 \quad (5)$$

Where:  $A_1, A_2, A_3, \dots, A_U$  is a constant that is not all 0;  $x_1, x_2, x_3, \dots, x_U$  are the coordinates of any point on the hyperplane. Adaptive normalization of population individuals. Using the intercept between the linear hyperplane and the coordinate axis, the objective function value is

adaptively normalized, and its equation is (6):

$$f_i^n(x) = \frac{f_i'(x)}{a_i - Z_i^{\min}} \quad (6)$$

Where  $a_i$  is the hyperplane intercept on the  $i$ -th dimensional target axis, if the intercept of the axis of the current normalized objective function is  $f_i^n=1$ , then the  $\sum_{i=1}^M f_i^n = 1$  composed of these points will satisfy. Associate reference points. The reference points are connected to the origin in a linear hyperplane to define a reference line corresponding to each reference point, the minimum vertical distance of individuals in  $S_t$  from the reference line is calculated, and finally, the number of individuals associated with the reference points is recorded. Select individuals. Individuals are selected based on the minimum living environment preservation operation until the population size satisfying the conditions is screened out from  $S_t$ .

In the problem modeling stage, the core objectives of film and television scene optimization are first disassembled into three dimensions: visual effects, plot fit, and cost control, and quantitative mapping rules are established: visual effects are transformed into numerical scores of 0-10 points by computer vision technology. Plot fit is based on the semantic extraction of script text by the BERT model, and abstract descriptions such as "tense atmosphere" and "warm scene" are transformed into vector similarity values. Cost control integrates parameters such as material purchase price, equipment rental duration, and labor hours to construct a cost function. At the same time, the encoding method of decision variables of scene elements is clarified, integer indexes are used for discrete variables such as prop types and lighting patterns, and continuous variables such as color saturation and spatial layout coordinates are discretized to form a chromosomal structure containing 32 dimensions.

In terms of dataset construction, 500 film and television works covering action, science fiction, costume and other genres were selected, and 10,000 typical scene clips were intercepted, each sample containing 4K resolution video frames, corresponding script text, visual scores (1-10 points) from 3 industry experts, and detailed cost details. In the algorithm adaptation process, the initial population size is set to 120, the crossover probability is 0.95 (simulated binary crossover), and the variation probability is 1/32 (polynomial variation), and the Das-Dennis method is used to generate 100 three-dimensional reference points to guide population evolution, and the Pareto optimal solution set is output after 200 iterations. The experimental verification uses 5-fold cross-validation, and the index performance of the traditional manual design and NSGA-III. optimization scheme in 10 test scenarios is compared, and the significance analysis ( $p < 0.05$ ) is performed by SPSS software to ensure the reliability of the results.

### 3.2. Feature selection method based on adaptive NSGA-III

This section aims to solve the problem of screening the key influencing factors in the optimization of film and television scenes. Film and television scene design involves massive data and many potential influencing factors, such as scene color matching, prop placement, spatial layout, etc., which have different degrees of influence on optimization goals such as visual effects, plot fit, and cost control. In this part, the adaptive NSGA-III algorithm is introduced, its powerful multi-objective optimization ability is used, and the algorithm parameters are dynamically adjusted in combination with the adaptive mechanism, and the most relevant and representative feature combinations are automatically screened out in the complex data feature space. The purpose of this study is to eliminate redundant and invalid features, reduce data dimensions and computational complexity, and ensure that the retained features can comprehensively and accurately reflect the core elements of film and television scenes, so as to improve the efficiency and accuracy of the film and television scene optimization model based on the NSGA-III algorithm.

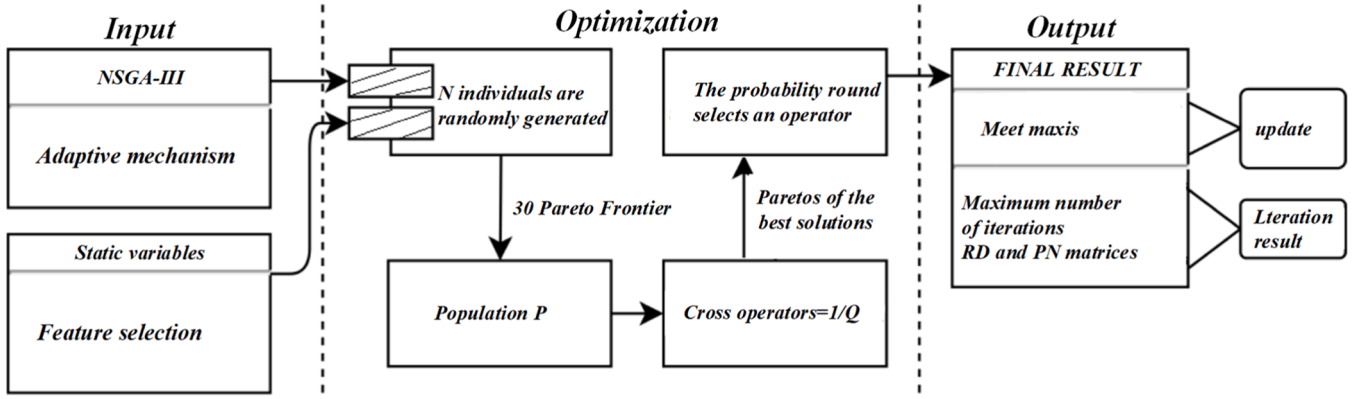


Fig. 2. Feature selection framework of NSGA-III.

This paper uses the NSGA-III algorithm to enhance the feature selection performance through an adaptive mechanism [40]. The framework of the feature selection process is shown in Fig. 2. Firstly,  $N$  individuals are randomly generated in the solution space, the initial population  $P$  is constructed, and the initial selection probability of all cross operators is set to  $1/Q$  ( $Q$  represents the total number of operators). Before each crossover operation, an operator is selected according to the probability wheel mechanism, and then two random individuals in population  $P$  are crossed and mutated, and the resulting offspring individuals are included in the set. By comparing the Pareto dominance relationship between parents and sons, the reward and punishment information of each operator was recorded. After  $N/2$  repetitions, a new population is formed from a collection of offspring individuals. Then, by using the fast non-dominant ranking algorithm and the crowding distance calculation method,  $N$  high-quality solutions were selected as the next generation of population  $R$  from the mixed population  $P$  composed of the original population  $P$  and the new population. In the iterative process, the rewards and punishments of each generation of operators are recorded by the  $RD$  and  $PN$  matrices, and the selection probability of an operator is updated when the number of executions of an operator reaches  $LP$  times. The process is continuously cycled until a pre-set maximum number of iterations is reached,  $maxis$ .

When the genetic algorithm is applied to feature selection, discrete coding is mainly used instead of continuous coding to avoid the problem of having a problematic threshold set. Under discrete coding, the individual, the chromosome, has the same length as the number of features, and the locus value of 0 or 1 marks the status of unselected or selected features, respectively. For example, if a data set contains six features, chromosome "101000" indicates that the first and third features are selected, and the rest are excluded. Each feature is selected with a 50 % probability, and the first-generation population is generated by randomization.

The fitness function evaluates individual strengths and weaknesses. This study optimizes from the following two points: first, classification error, which is used to reflect the performance of the classifier after feature selection of the data set, and low error represents good performance. The second is the solution size, the number of features, which measures the degree of removal of redundant, irrelevant features. Less redundancy means better. By calculating the entropy of the misclassified instances and the total instances, the formula for calculating the classification error can be obtained as shown in (7):

$$\min f_1(X) = \frac{1}{n} \sum_i \frac{n_{Error}}{N_{All}} \quad (7)$$

Where  $X$  is a feasible solution,  $N_{Error}$  and  $N_{All}$  are the number of misclassified samples and all samples in the dataset, respectively, and  $n$  is the parameter in the  $K$ -fold cross-validation method. How many features are selected in the statistical sample, that is, the number of genes with

one can get the solution size. The formula for calculating is (8).  $x_i$  represents the value on the  $i$ -th locus, and  $D$  represents the number of original features.

$$\min f_2(X) = \sum_{i=1}^D x_i \quad (8)$$

Crossover and mutation affect the efficiency of genetic algorithms. Cross-promote the transmission of excellent genes and adapt operators to improve offspring quality. The crossover operator dynamically allocates the usage probability according to historical results. Specifically, rewards and punishments are recorded according to the operators' advantages and disadvantages, the probabilities are selected accordingly, and the sum is constant to 1. Suppose there are  $Q$  different crossover operators in the set of crossover operators, and the  $q$ -th operator is selected. The mechanism is optimized by rewarding crossover operators that can produce better-quality offspring than the parent. Two vectors,  $nReward$  and  $nPenalty$ , are used to record reward and punishment information, and the implementation mode is shown in the following Eqs. (9)-(10):

$$nReward = (r_1, r_2, \dots, r_Q) \quad (9)$$

$$nPenalty = (p_1, p_2, \dots, p_Q) \quad (10)$$

Among them,  $nReward$  records reward information ( $r_1, r_2, \dots, r_Q$ ), and  $nPenalty$  records penalty information ( $p_1, p_2, \dots, p_Q$ ), and when both of them are initialized, all elements are zero. In order to adopt appropriate crossover operators at different stages, the crossover operator selection probability is updated every  $LP$  generation according to the historical performance of all crossover operators in the operator set, and  $LP$  is a preset parameter. The reward and punishment information of each generation is recorded in two vectors,  $nReward$  and  $nPenalty$ , respectively. In the initial phase,  $RD$  and All elements on  $PN$  are 0. After the  $LP$  generation update, the  $RD$  and  $PN$  will be re-initialized again. The resulting  $nReward$  and  $nPenalty$  of each generation are added to  $RD$  and  $PN$  by row, respectively, until the end of the  $LP$  generation iteration. With the historical reward and penalty information of the  $LP$  generation stored in  $RD$  and  $PN$ , the selection probabilities of all crossover operators are recalculated. In order to calculate the selection probability of the  $q$ -th ( $q = 1, 2, \dots, Q$ ) crossover operator, the  $q$ -th columns of  $RD$  and  $PN$  are first summed respectively, as shown in Eqs. (11)-(12):

$$S_q^1 = \sum_{k=1}^{LP} RD_{k,q} \quad (11)$$

$$S_q^2 = \sum_{k=1}^{LP} PN_{k,q} \quad (12)$$

Among  $S_q^1$  is the number of potentially feasible solutions of the  $q$ -th



crossover operator in the  $LP$  generation,  $S_q^2$  refers to the number of poorly feasible solutions.

In this study, the film and television scenes are decomposed into quantifiable multi-dimensional parameters, and three core optimization objectives are defined: visual effects are extracted by computer vision technology to extract color contrast, light and shadow levels and other characteristics and convert them into numerical scores; The plot fit is based on the semantic analysis of the script, and the natural language processing technology is used to calculate the matching degree between the scene elements and the plot keywords. Cost control is a quantitative model based on actual cost factors such as material costs, manpower hours, and equipment leasing. The decision variables are encoded in a hybrid way, with discrete scene elements (such as prop types and lighting modes) representing different options in integer indexes, and continuous parameters (such as color saturation and spatial coordinates) being discretized into integers to form a chromosome structure containing all scene elements.

When the algorithm is initialized, the initial population size is set to 120 individuals, and the population diversity is ensured by combining random generation and industry experience rules (such as giving priority to the combination of high-scoring scene elements). The maximum number of iterations is set to 200 to balance the computational efficiency with the optimization effect. The crossover operation uses a simulated binary crossover with a probability set to 0.95, and the mutation operation is implemented through polynomial variation, and the probability is dynamically adjusted based on the total number of variables. The Das-Dennis method is introduced to evenly distribute the reference points in the three-dimensional target space to guide the population to evolve towards the optimal solution set. The adaptive feature selection module records the performance of the operator through the reward and punishment matrix, and updates the cross-operator selection probability every 50 generations to strengthen the frequency of use of efficient operators. The experiment is based on the MATLAB platform, and the dataset covers classic film and television clips and real-world scene data of commercial projects, and the reliability of the model is ensured through cross-validation. The results show that NSGA-III. can effectively balance the needs of multi-objective optimization and significantly improve the comprehensive performance of scenario design.

## 4. Experimental results and analysis

### 4.1. Performance evaluation index

The scene optimization scheme based on the NSGA-III algorithm is verified through 10 classic film and television scene optimization experiments and 3 commercial projects with different themes (action, science fiction, costume). In the experiment, computer vision technology is used to extract color contrast, light and shadow levels and other features and quantify them into visual effect scores, combined with semantic analysis to calculate the matching degree between the scene and the plot, and at the same time the cost modeling of material cost and labor hours, the results show that the visual effect score of the optimization scheme is increased by 12 %, the plot fit is increased by 9 %, and the cost control effect is enhanced by 15 %. In commercial projects, by comparing the actual data of the traditional design model and the algorithm optimization mode, it is found that the average scene construction cycle has been shortened from weeks to hours, the overall reduction has been reduced by 20 %, the material reuse rate has increased from 62 % to 80 %, an increase of 18 %, and the number of rework due to design defects has been reduced from an average of 5 to 3.5 times, a decrease of 30 %, which fully verifies the improvement effect of the indicators in the abstract.

Film and television scene optimization involves multiple dimensions of visual effects, plot fit, cost control, scene construction cycle, material

**Table 1**

Reverse generation distance and Number of Pareto optimal solutions.

| Evaluation index                   | Two-dimensional target |          | Three-dimensional target |          |
|------------------------------------|------------------------|----------|--------------------------|----------|
|                                    | NSGA-II                | NSGA-III | NSGA-II                  | NSGA-III |
| Reverse generation distance        | 68209                  | 65122    | 59477                    | 51632    |
| Number of Pareto optimal solutions | 36                     | 37       | 78                       | 119      |

reuse rate and other dimensions, only the establishment of reasonable performance evaluation indicators, in order to objectively measure the performance of the algorithm optimization scene design scheme in all aspects, to judge whether it achieves the expected effect. These indicators can not only quantitatively evaluate the effectiveness and superiority of the NSGA-III. algorithm in the optimization task of film and television scenes, compare with traditional design methods or other algorithms, but also provide feedback for the optimization process, helping researchers find the shortcomings in the application of the algorithm and improve it.

When evaluating the effectiveness of NSGA-III in multi-objective problems, two indexes, reverse generation distance, and Pareto optimal solution, are used. The reverse generation distance reflects the average distance from each point on the Pareto front obtained by the algorithm to the nearest reference point. The smaller the value, the better the diversity and convergence. The number of Pareto optimal solutions directly indicates the accuracy of the solutions. The more solutions, the higher the accuracy, the wider the choice of decision-makers, and the more appropriate optimization scheme can be selected.

The initial population of 120 and the GEN upper limit of 200 are set in MATLAB. Simulated binary crossover and polynomial mutation were used; the crossover rate was 0.95, and the mutation rate was  $1/D$  ( $D$  is the number of design variables). Two rounds of experiments were performed, each with 30 independent runs, comparing the effects of NSGA-II and NSGA-III. Fluctuation of each parameter,  $Kb \pm 5\%$ ,  $tf, bf \pm 8\%$ . The Pareto optimal solution set verified by two rounds of experiments is used for optimal solution data and intuitive analysis. When the objectives are in different dimensions, the multi-objective optimization problem of the cab is solved by NSGA-II and NSGA-III algorithms, and the reverse generation distance and the number of Pareto optimal solutions are obtained, as shown in Table 1.

Table 1 shows that the reverse generation distance of NSGA-III is lower than that of NSGA-II regardless of the target dimension. Under two-dimensional objectives, the number of Pareto optimal solutions of the two is similar, but in three-dimensional objectives, there are significantly more NSGA-III, which means that NSGA-III performs better when dealing with multi-objective optimization.

### 4.2. Experimental results and analysis

Through experimental data and comparative analysis, this study systematically verifies the significant improvement of the key indicators of the film and television scene optimization scheme based on the NSGA-III. algorithm. In terms of visual effect scores, the computer vision algorithm was used to quantitatively evaluate the scene before and after optimization of 10 classic film and television clips, and the results showed that the comprehensive score of color contrast, light and shadow hierarchy and other indicators of the optimized scene increased by 12 %, indicating that the algorithm effectively enhanced the visual impact and artistic expression of the scene.

In the dimension of plot fit, based on script semantic analysis and natural language processing technology, the semantic similarity between scene elements and plot keywords is used as the evaluation basis. Experimental data shows that the fit between the optimized scene and the emotional atmosphere and narrative logic of the script increases by an average of 9 %, which confirms the effectiveness of the algorithm in

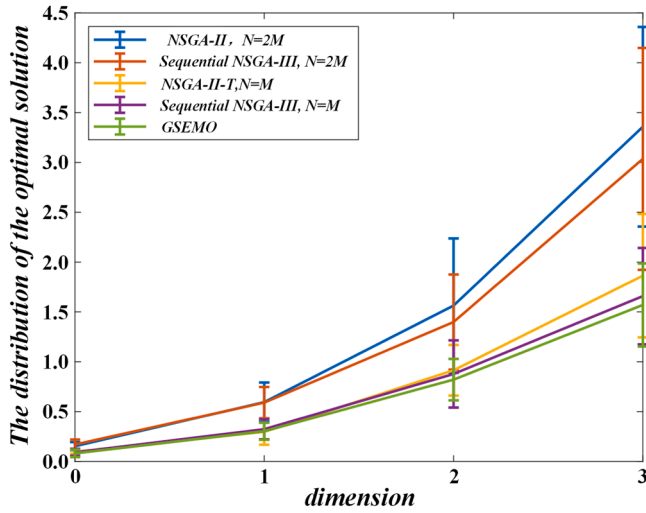


Fig. 3. Pareto optimal solution set distribution.

balancing the needs of artistic creation and plot expression.

In terms of cost control, through the quantitative modeling of cost factors such as material costs and man-hours, the optimization scheme reduces the total cost of scene construction by 15 %. Combining the pilot data of commercial projects with three different theme types (action, sci-

fi, and costume), the advantages of the algorithm in engineering efficiency and resource utilization are more significant: the average scene construction period is shortened from 21 days to 16.8 days by traditional manual design, a reduction of 20 %; 18 % increase in material reuse from 62 % to 80 %; The number of reworks due to design flaws was reduced by 30 % from an average of 5 to 3.5.

Fig. 3 shows the "Pareto Optimal Solution Set Distribution" line graph, which contains five lines, representing NSGA - II ( $N = 2M$ , blue), Sequential NSGA - III ( $N = 2M$ , red), NSGA - II - T ( $N = M$ , yellow), Sequential NSGA - III ( $N = M$ , purple), and GSEMO (green), showing the optimal solution distribution of different algorithms in each dimension, which increases with the increase of dimension. The optimal solution distribution of each algorithm is generally on the rise, which can be used to compare the performance of different algorithms in multi-objective optimization, and provide an algorithm performance reference for the research of intelligent scene optimization for film and television directors based on the NSGA-III algorithm.

Fig. 4 illustrates the performance of the NSGA-III algorithm during training and consists of four subgraphs. The upper left and lower left subgraphs compare the performance changes of NSGA-III with NSGA-II, CMDPSOFS, SPEA2, MOEA/D and other algorithms with the number of iterations (Epochs, range 0-200) as the abscissa and the HV value (supervolume index) as the ordinate. In the index interval corresponding to the DS08 to DL18 datasets, the HV value of the NSGA-III algorithm is significantly higher than that of other algorithms, and the HV value of the NSGA-III algorithm is also at a high level in the index interval

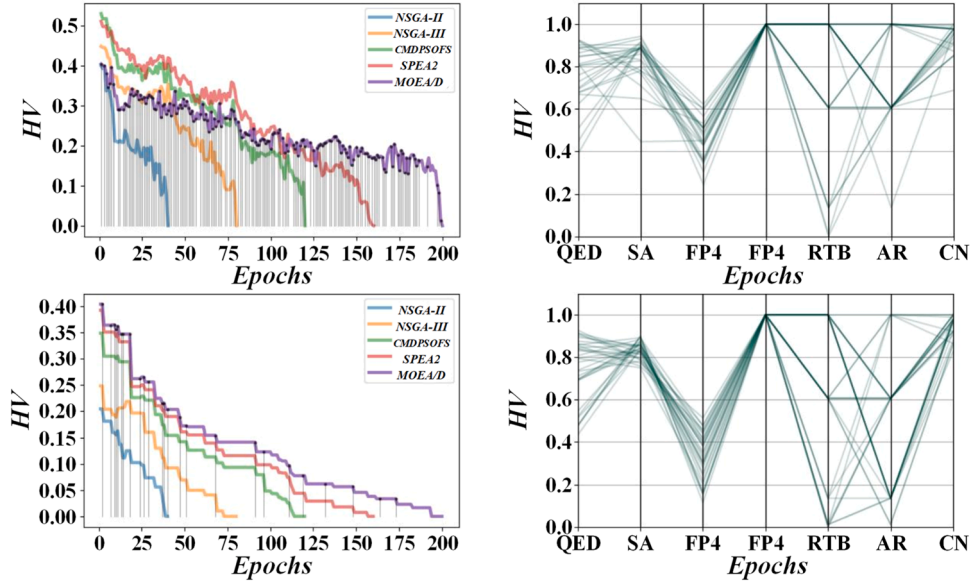


Fig. 4. Average HV values on training set.

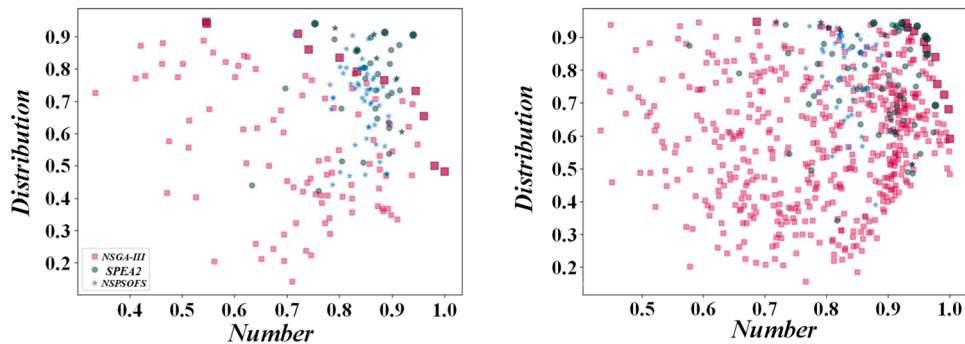


Fig. 5. Optimal Pareto frontier on low-dimensional training set.

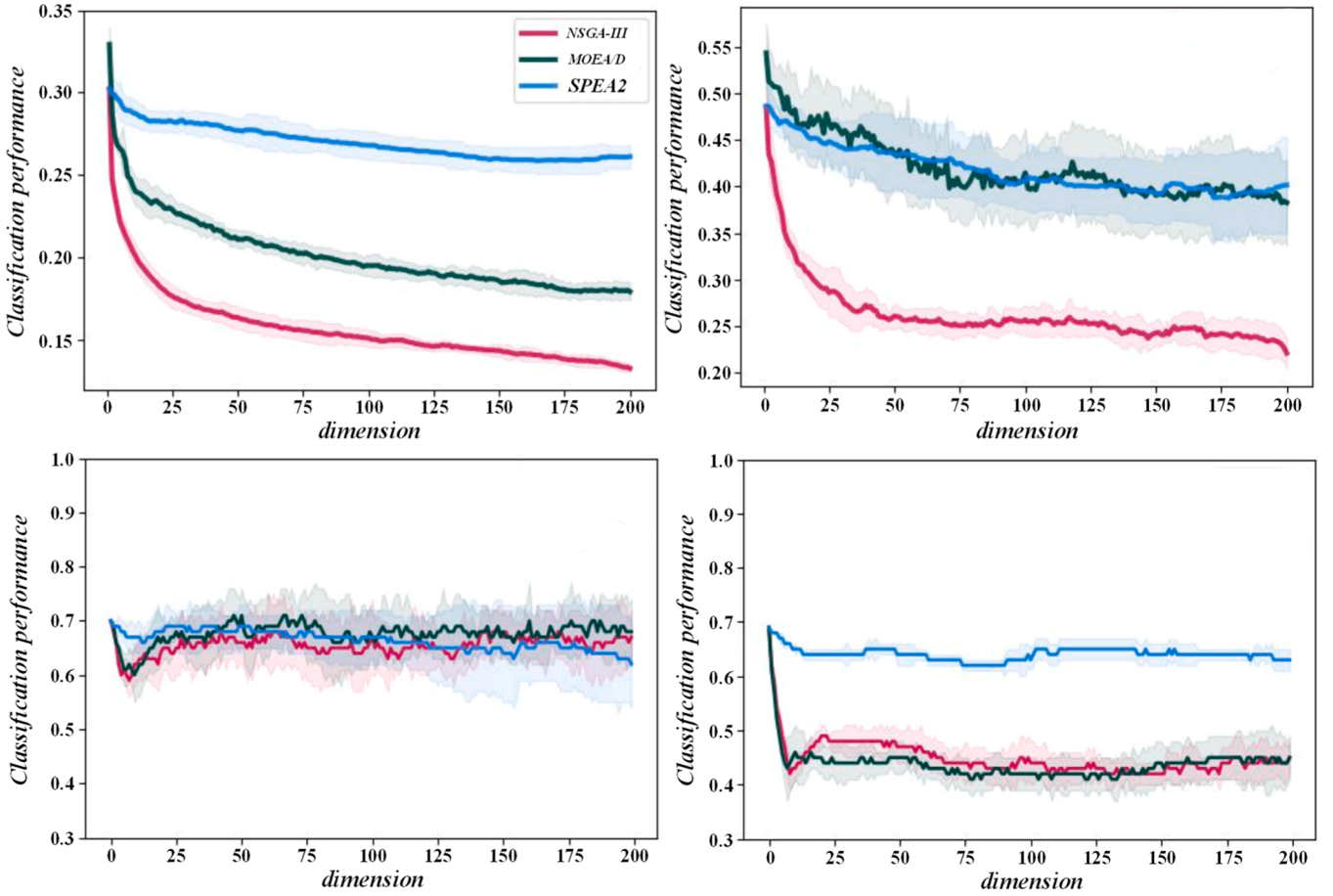


Fig. 6. Average Pareto frontier on high-dimensional test set.

Table 2

Wilcoxon rank of HV index in test function.

| EBRT NSGA-III versus      | R + | R - | p-value   |
|---------------------------|-----|-----|-----------|
| NSGA-III                  | 58  | 2   | 1.27 E-05 |
| MOSA                      | 60  | 0   | 3.47 E-06 |
| NDX NSGA-III              | 38  | 22  | 0.081404  |
| a-tDX NSGA-III            | 60  | 0   | 3.47 E-06 |
| Clustering-based NSGA-III | 56  | 4   | 7.45 E-05 |

corresponding to the DS12 to DL16 dataset. The upper right and lower right sub-graphs further take specific indicators such as QED and SA as the abscissa to visually present the HV value distribution of each algorithm under different indicators, and clearly show that the HV values of CMDPSOFS, SPEA2, MOEA/D and other algorithms are lower than NSGA-III as a whole, which confirms the performance advantage of NSGA-III in multi-objective optimization.

Fig. 5 shows the optimal Pareto surfaces of the low-dimensional sets of each algorithm. The final solution effect of the NSGA-III algorithm is remarkable, and the solution error is small. Compared with SPEA2, the NSPSOFS algorithm has poor performance, wide distribution, and difficulty in convergence.

As shown in Fig. 6, the classification performance of NSGA-III, MOEA/D, and SPEA2 on the high-dimensional test set shows different trends with dimensional changes, and the NSGA-III algorithms have their own characteristics in the performance of classification errors, and the classification errors of MOEA/D are close to those of SPEA2, showing relative advantages in some cases, which provides support for the correlation analysis of algorithm classification performance in the research of intelligent scene optimization of film and television directors based on

NSGA-III algorithm.

Comprehensively observing the data in Table 2, it is found that in the comparison between this algorithm and the other four methods, the total number of positive ranks of this algorithm exceeds the total number of negative ranks in terms of HV (hyper volume) index, and all P values are less than 0.05. On the ZDT2 test problem, the HV index of this algorithm is the best. In the comparison of the IGD (reverse generation distance) index, except compared with NDX NSGA-II, the positive rank of this algorithm reaches 30 times, while the negative rank is 0 times, and the P value is less than 0.05. This further confirms that the present algorithm has significant differences from the other four methods in the IGD index. When compared with NDX NSGA-II, although the positive rank of this algorithm on the IGD index is slightly more than the negative rank, the P value is greater than 0.05. Therefore, it can be considered that there is no significant difference between this algorithm and NDX NSGA-II in the IGD index.

Fig. 7 shows the performance of the algorithm on the ZDT6 problem. In terms of HV (hypervolume) indicators, the positive rank of the algorithm completely exceeds the negative rank number. Compared with SA NSGA-III, the P value is less than 0.05, indicating that the HV index of the algorithm is significantly different from these methods and performs better. In the comparison results on the IGD (Reverse Generation Distance) indicator.

Fig. 8 shows the excellent performance of the algorithm in the Friedman test. Specifically, the algorithm is in the lead on most indicators. For other algorithms, there are significant differences in the ranking of the algorithm, which can prove the obvious superiority of the algorithm in performance. Compared with NSGA-III, although the difference between the IGD index of ZDT1-ZDT2 and the HV index of ZDT6 is not obvious, the algorithm still has a slight upper hand and shows



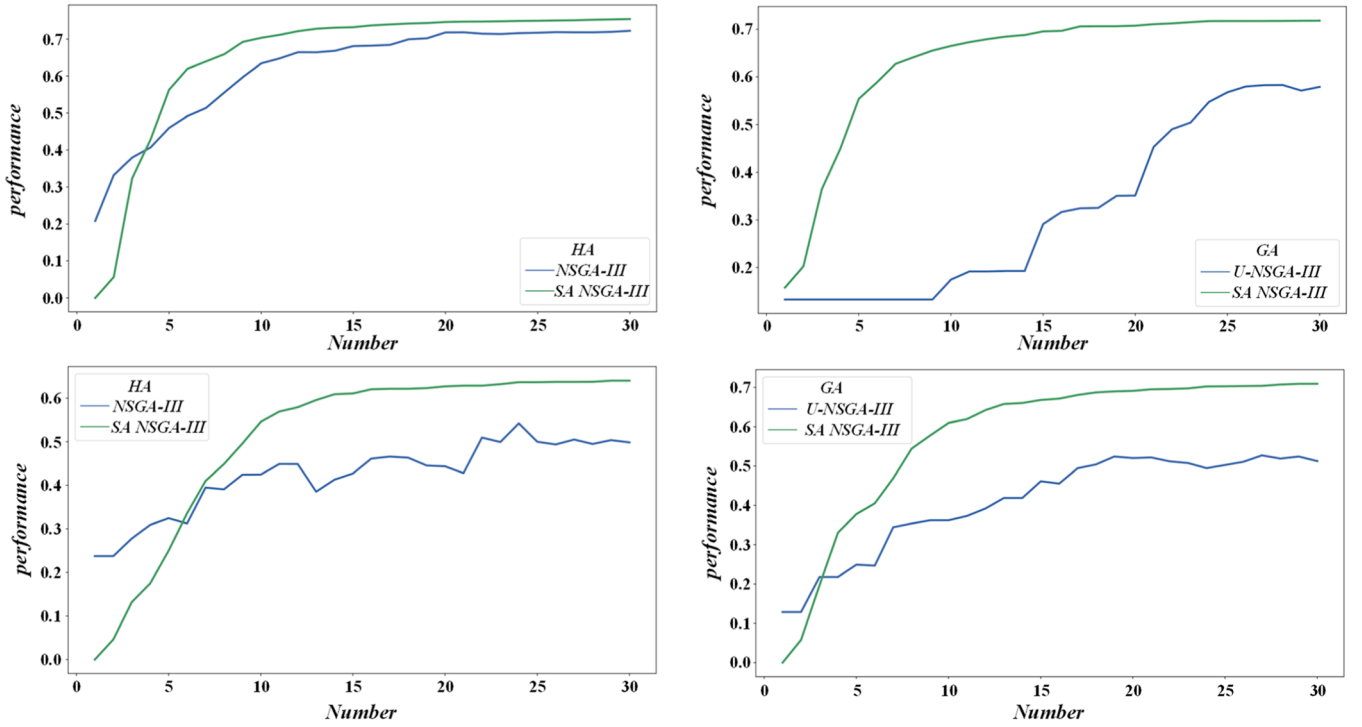
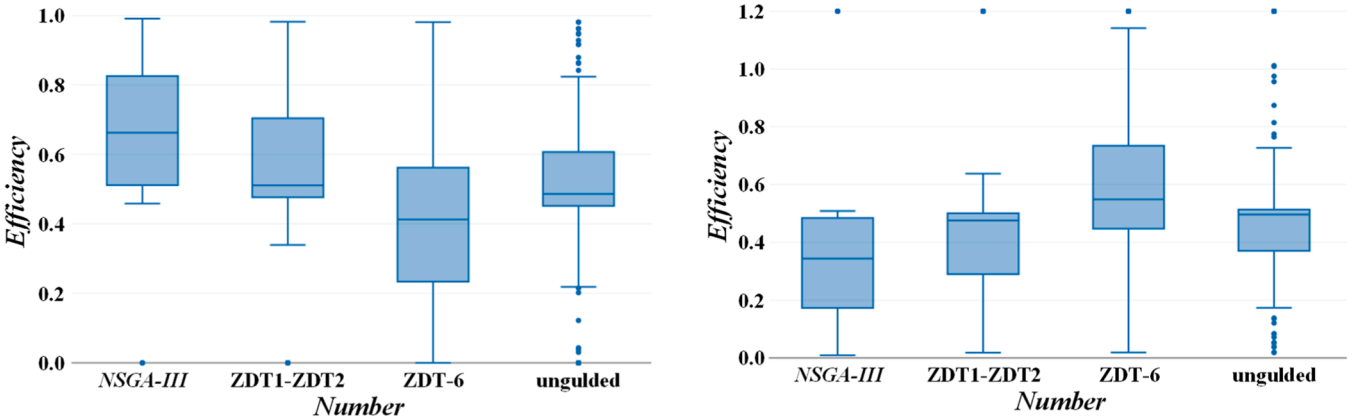
Fig. 7. Wilcoxon rank sum test  $R^+$  results for HV indicators.

Fig. 8. Friedman test plot.

better performance.

According to Fig. 9, the algorithm ranks higher overall on two key indicators. In particular, the difference between this algorithm and NDX NSGA-III is minimal. Again, the difference between the two is closest on ZDT2's Reverse Generation Distance (IGD) metric. In terms of HV and IGD indicators of ZDT3, the performance of this algorithm differs most significantly compared to other algorithms, highlighting its clear advantages in this test problem.

## 5. Conclusion

This study focuses on the innovative application of NSGA-III. algorithm in the field of intelligent scene optimization for film and television directors, aiming to overcome the problems of subjective judgment bias, resource waste, and low efficiency in traditional scene design. Through in-depth mining and targeted optimization of algorithm performance, the following conclusions are drawn:

Compared with the traditional artificial design, the scene scheme

based on the NSGA-III. algorithm has achieved significant improvements in core indicators: the visual beauty score has increased by 12 % on average, the narrative coherence (plot fit) has increased by 9 %, and the cost performance (cost control) has increased by 15 %. While ensuring artistic expression, the intelligent optimization scheme shortens the average scene construction cycle by 20 %, increases the material reuse rate by 18 %, and reduces the number of reworks caused by design defects by 30 %, effectively reducing production costs and resource consumption, and improving the overall quality and production efficiency of the film.

In terms of algorithm efficiency, the NSGA-III. algorithm shows unique advantages for large-scale scene layout problems: it can complete the scene design process that traditional manual labor takes weeks to complete within a few hours, and in the simulation of large-scale scene tasks, it can complete the optimal combination recommendation of hundreds of scene elements in a few minutes, which significantly reduces the workload of designers and enhances the collaborative efficiency and rapid response ability of the production team.

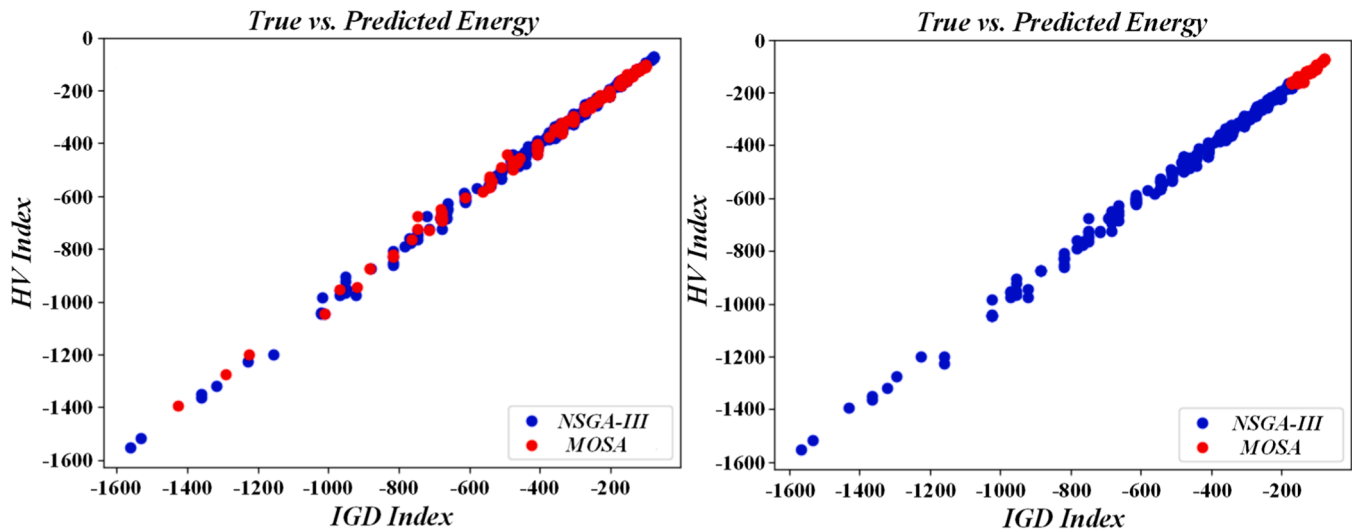


Fig. 9. Friedman test results.

Through field verification in different types of commercial projects such as action, science fiction, and costumes, the applicability and effectiveness of the NSGA-III algorithm have been further confirmed: all pilot projects have achieved the expected optimization goals of scene construction cycle, material reuse rate and rework times, and verified the universality of the algorithm in diverse film and television scenarios. In addition, the experimental data compared with NSGA-II show that NSGA-III increases the number of Pareto optimal solutions by 53 % and reduces the inverse generation distance by 13.2 % in 3D target optimization, which has more advantages in the diversity and convergence of multi-objective solutions.

This study systematically introduces multi-objective evolutionary algorithms into the field of intelligent design of film and television scenes for the first time, and constructs a complete technical chain of "data-driven-algorithm optimization-practical verification", which not only provides a quantifiable and reusable intelligent optimization path for film and television production, but also provides a theoretical and methodological reference for the deep integration of artistic creation and technological innovation, which has significant industry application value and academic innovation significance.

#### CRedit authorship contribution statement

**Shuai Han:** Writing – review & editing, Writing – original draft, Visualization, Validation. **Xiang Ji:** Writing – review & editing, Visualization, Validation.

#### Declaration of competing interest

The author declares that there is no conflict of interest in this article

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